

Supply Chain Disruption Assistant & Fleet Utilisation Optimizer

Team : The Outliers



Supplier

Port

Warehouse

Fleet

Customer



Problem Statement

01 Disruptions

Weather events, port strikes, and geopolitical crises impact multiple shipments at once.

02 Shipment Visibility

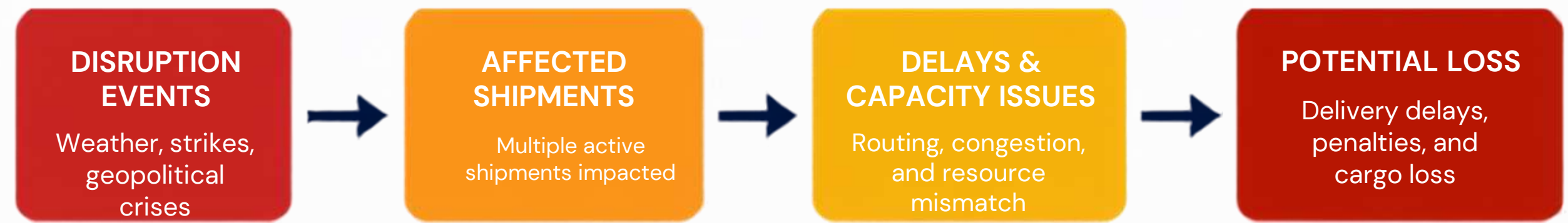
operators must quickly identify which active shipments are affected.

03 Fleet Imbalance

Idle trucks, containers, and vessels while other routes face heavy congestion.

04 Cold-Chain Risk

Temperature excursions are often detected too late, leading to cargo spoilage and loss.



Current Challenges

OPERATIONAL CHALLENGES

- Scattered disruption information
- Slow identification of affected shipments
- Manual routing and carrier decisions
- Underutilised idle fleet capacity
- Difficult continuous cold-chain monitoring

CONSEQUENCES

- Delayed response to disruptions
- Capacity imbalances persist
- Higher operational costs
- Late detection of temperature risks

MULTIPLE DATA SOURCES

Information from many systems and partners



MANUAL ANALYSIS

Fragmented review and spreadsheet-driven checks

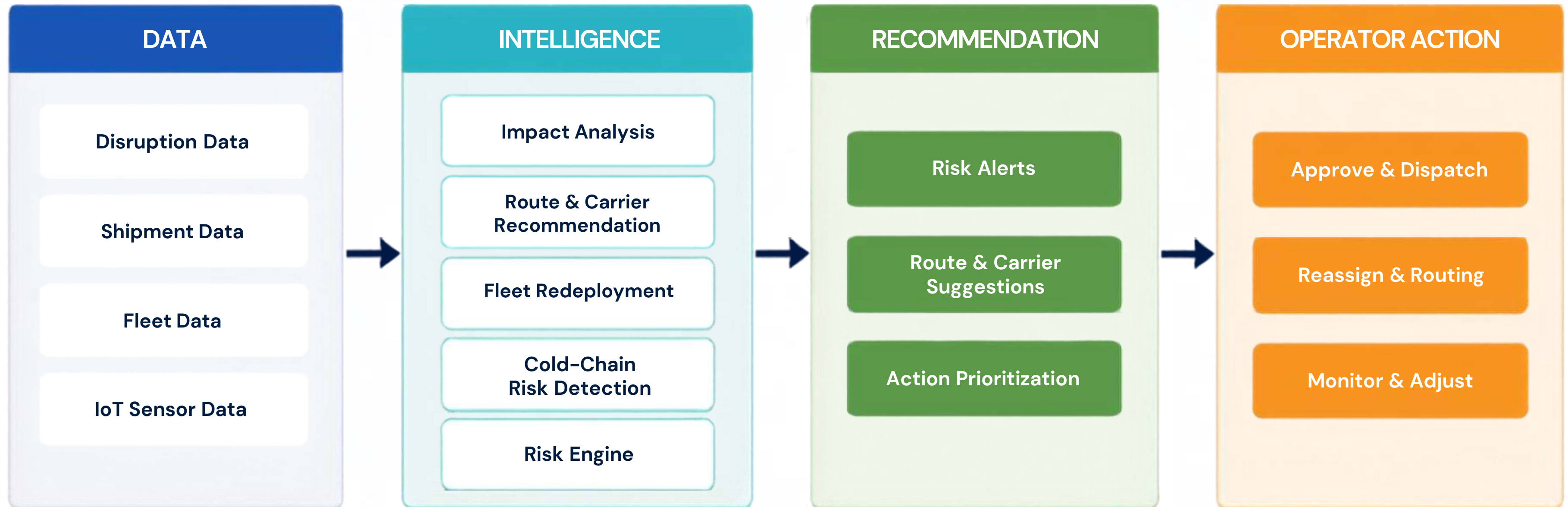


DELAYED DECISION

Slower actions lead to missed opportunities and disruptions

Proposed Solution

AI-Assisted Supply Chain Decision Platform



The platform integrates disruption, shipment, fleet, and IoT data to identify risks and recommend operational actions.

Key Features



01

AI-Powered Disruption Detection

Maps active
disruptions to
affected shipments.



02

Route & Carrier Recommendation

Evaluates and suggests
alternative
routes/carriers.



03

Fleet Utilisation Optimization

Identifies idle assets
and suggests
redeployment.



04

Cold-Chain Monitoring

Analyzes IoT
temperature logs to
detect excursions.

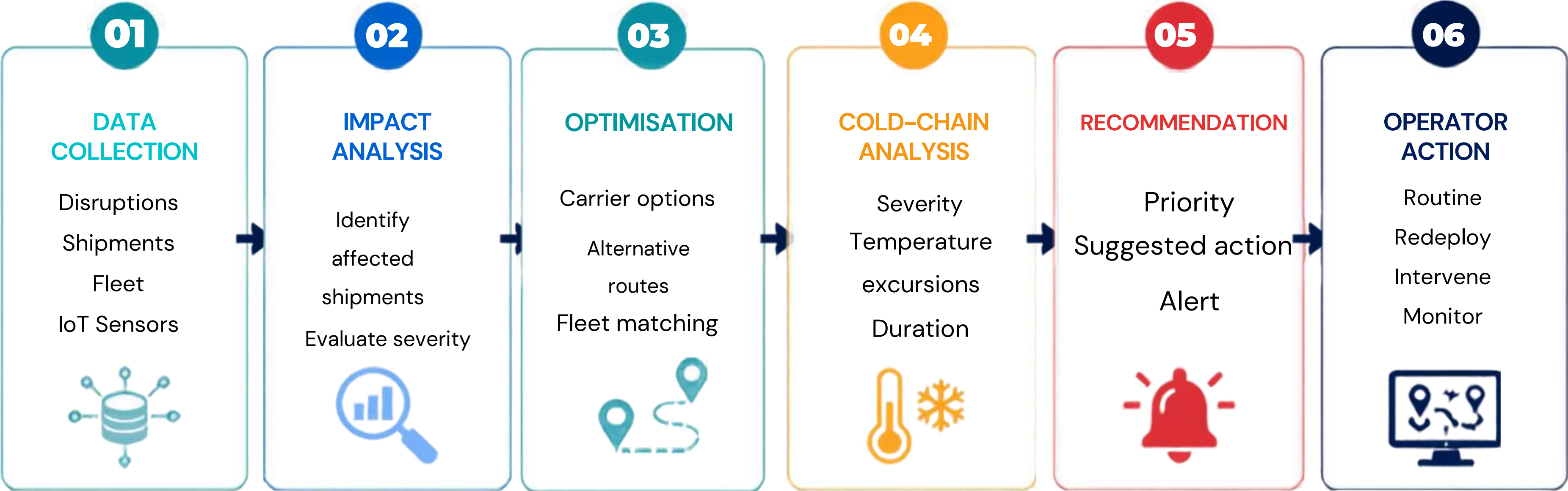


05

Risk & Severity Assessment

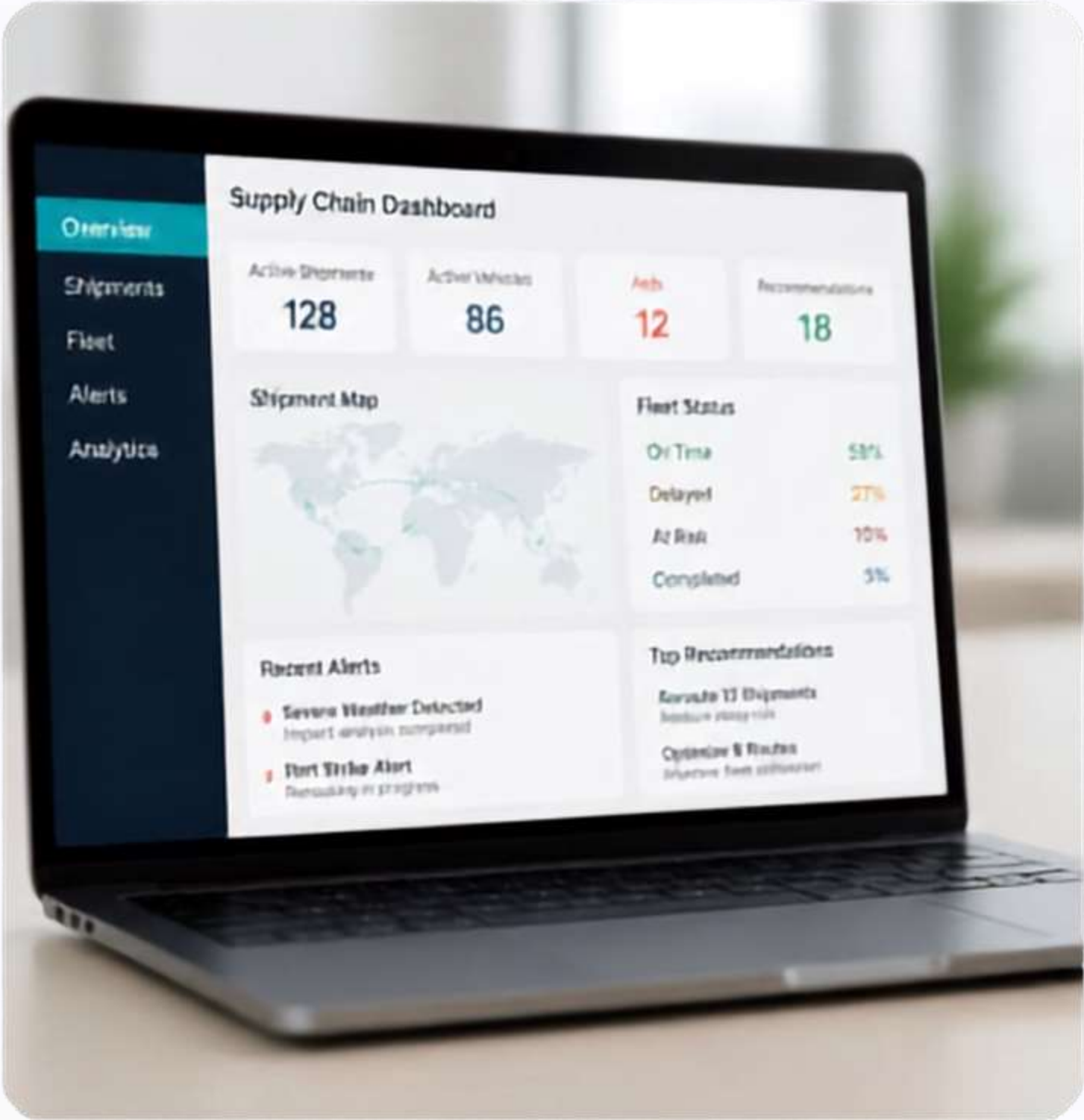
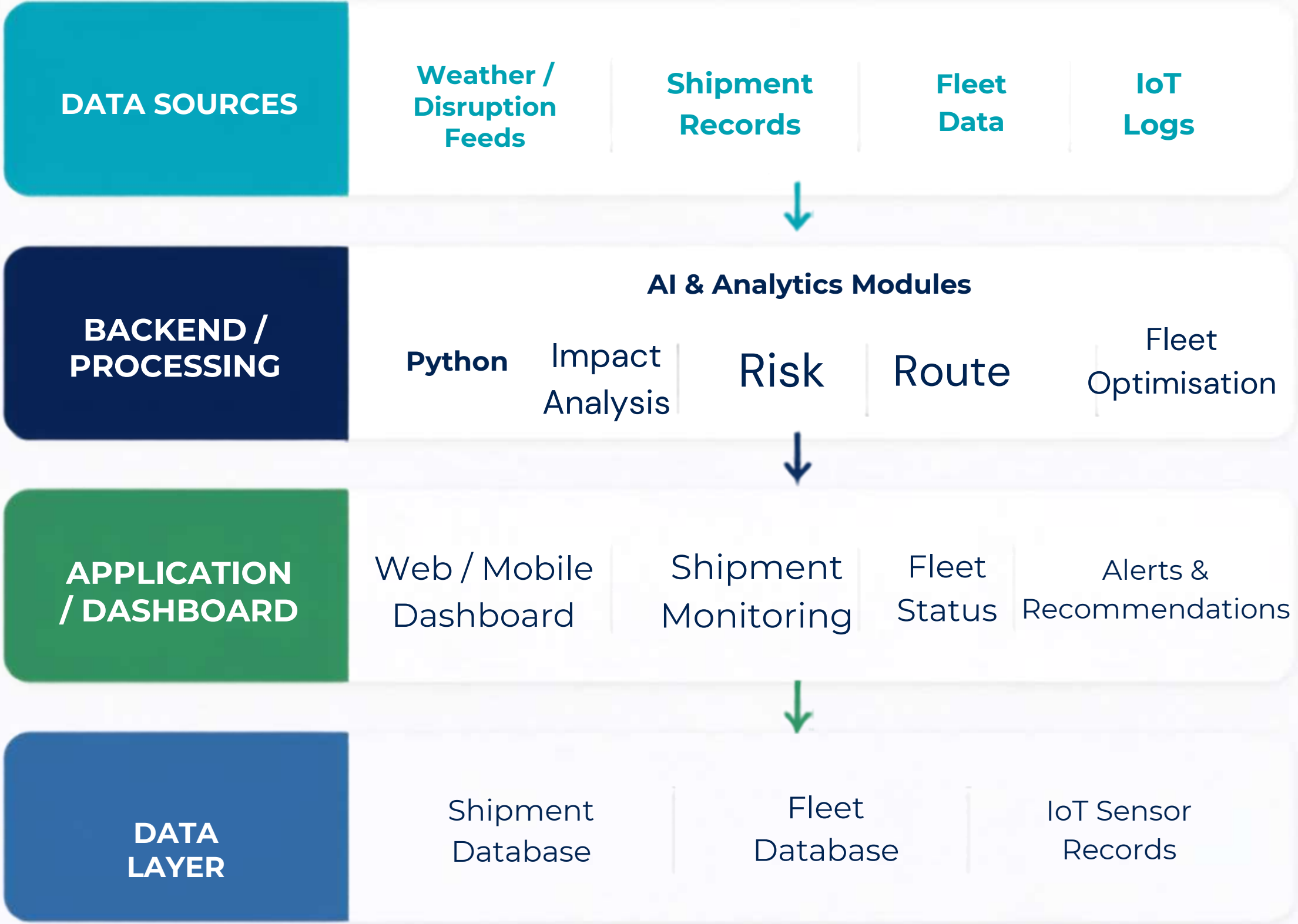
Classifies incidents
and prioritizes
attention.

System Workflow



Data → **Analysis** → **Recommendation** → **Action**

Technology Stack & Implementation



Innovation & Feasibility

INNOVATION

Integrated Decision Support

connects disruption, shipment, fleet, and cold-chain data

Action-Oriented Analysis

recommends decisions instead of only detecting problems

Combined Optimisation

considers shipment impact along with fleet capacity

Proactive Cold-Chain Monitoring

detects temperature excursions early to prevent spoilage



FEASIBILITY

Modular Architecture

supports independent services and seamless integration

Leverages Existing Data

uses shipment, fleet, disruption, and IoT data sources

Scalable Processing

built to handle large volumes of shipments and assets

Human-in-the-Loop Assurance

operators validate and execute recommended decisions

Expected Impact & Future Scope

01

Faster Response

Early identification of affected shipments

02

Better Fleet Utilisation

Reduced idle capacity

03

Improved Shipment Continuity

Quicker routing and carrier decisions

04

Reduced Cold-Chain Risk

Earlier temperature excursion detection

Future Scope

Real-time disruption feeds

Predictive disruption modelling

Live GPS fleet tracking

Automated carrier comparison

Advanced demand forecasting

Digital twin / simulation

Integration with enterprise logistics systems

